Does Kernel-Owned Networking Make GPU Serving Cheaper?

Lessons from BPFusion

Yusheng Zheng
UC Santa Cruz
Santa Cruz, CA, USA
yzhen165@ucsc.edu

ABSTRACT

A tempting design for small-model GPU serving removes the userspace request loop entirely: eBPF frames and admits requests arriving on ordinary Linux TCP sockets, a resident GPU dispatcher executes CUDA graphs, and a BPF program driving io_uring sends the replies. We built such a system, BPFusion, on stock Linux 7.3 without a kernel module, and compared it with an efficient C++ epoll server that drives the same GPU executor. Across LLM serving with Qwen2.5-0.5B and high-rate cuBLAS/cuFFT requests (more than 3.4 million measured requests in the final experiments), moving the request loop into the kernel did not reduce latency or total CPU. Throughput was identical, and per-request CPU moved from user to system time (11.3+26.2 μs versus 2.8+35.2 μs). The large improvements we found, a 15× lower completion tail from adding GPU slots and a 17× lower token-gap tail from disabling Nagle's algorithm, apply equally to both transports. The kernel TCP stack and completion coordination dominate per-request cost, and both remain on the kernel-owned path.

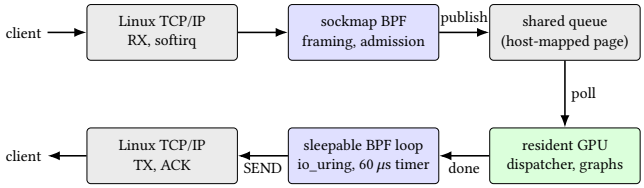


Figure 1: Steady-state path of the kernel-owned transport. Blue boxes are eBPF, gray boxes are native kernel code, green is GPU. Userspace appears only at startup. The comparator replaces both blue boxes with one C++ epoll thread and keeps everything else.

1 INTRODUCTION

Serving a small model, or a short GPU kernel, at a high request rate makes host overhead visible. Each request still crosses the network stack, wakes a server thread, is copied into user memory, is queued for the GPU, and returns through another send. When the computation itself takes microseconds to a few milliseconds, this shepherding can be a large share of the work.

Two recent Linux capabilities suggest removing the shepherd. Sockmap BPF programs can parse and steer data on TCP sockets inside the kernel, and Linux 7.3 lets a sleepable BPF program submit and complete io_uring operations itself. Combined with a persistent GPU dispatcher that launches captured CUDA graphs without host involvement, a server needs userspace only at startup: after initialization no user thread reads, schedules or sends a request. Intuitively this should save CPU and shorten tails, because it skips system calls, user copies and thread wakeups.

We built this design as BPFusion and asked a narrow question: *compared with a well-engineered userspace server that drives the same GPU executor, does kernel-owned networking reduce CPU or latency?* The answer on our testbed is no. This paper reports:

- a working kernel-owned TCP-to-GPU serving path on stock Linux, validated with real models and unchanged cuBLAS/cuFFT graphs (§2);
- a controlled comparison that changes only the transport, plus an external vLLM reference and explicit CPU accounting (§3);
- a negative result for throughput, tail latency and total CPU, together with an explanation of where the cost goes (§4, §5);
- transport-independent fixes that mattered far more than the transport (§4.3).

2 BPFUSION

At startup a userspace process creates the listening socket, captures the caller's CUDA graphs, maps a shared page that both the CPU and GPU can access, attaches the BPF programs, and launches a resident GPU dispatcher. It then stays alive only to own these allocations and to handle shutdown.

Figure 1 shows the steady state. Linux TCP reassembles bytes as usual. A sockmap BPF program frames each request (a 24-byte header followed by 32-bit words, optionally wrapped in a Content-Length HTTP POST), validates it, and publishes it into a FIFO in the shared page. The GPU dispatcher reads published requests with system-scope acquire loads, launches the matching caller-owned graph, and publishes outputs. Two compute adapters exist: an opaque one-shot adapter for unchanged cuBLAS and cuFFT graphs, and a generation adapter that replays upstream vLLM or Hugging Face prefill and decode graphs for Qwen2.5-0.5B and SmolLM2-135M. A sleepable BPF program attached to io_uring consumes completions, submits SEND operations for finished outputs, preserves reply ordering, and reclaims connections. While work is active it waits for the next completion event or a 60 μs timer.

Some parts of the path remain native kernel code, and they matter for the results: TCP/IP processing with its interrupts and softirqs, the io_uring core and its io-wq workers, and the thread context in which the BPF loop runs, which belongs to the service process.

Table 1: Qwen2.5-0.5B at 192 requests/s. Top: transport comparison with 16 GPU slots. Bottom: a separate run that raises common capacity to 32 slots (native transport). CPU is process-group CPU per request.

Path	tokens/s	P99 (ms)	CPU (ms)
Kernel-owned, 16 slots	11,645	3,304	0.835
Native epoll, 16 slots	11,645	3,314	0.812
Official vLLM	12,245	611	8.545
Native epoll, 16 slots	11,640	3,337	0.751
Native epoll, 32 slots	12,265	217	0.523
Official vLLM	12,249	406	8.327

3 METHODOLOGY

Baselines. The main comparator, *native*, uses the same queue, the same GPU dispatcher and the same graphs, but replaces the two BPF components with a C++ thread that uses timed epoll and nonblocking reads and writes. Only the transport placement differs. Official vLLM 0.31 [8] with its optimized defaults serves as an external reference for the LLM workload; its HTTP API, scheduler and KV management differ, so it measures the whole serving design rather than the transport.

Testbed. The server has an Intel Core Ultra 9 285K (24 logical CPUs), an RTX 5090 and Linux 7.3-rc3. Load comes from a client on a separate host. Every condition runs in five paired process blocks with counterbalanced order; we report means of block statistics with nominal paired 95% intervals.

Workloads. LLM: Qwen2.5-0.5B in FP16, short prompts, 64 output tokens, 192 requests/s for 60 s (12,288 offered tokens/s). *Small graphs*: an 8×8 FP32 cuBLAS GEMM or a 32-point complex64 cuFFT, repeated 1, 16 or 256 times inside a graph, with 280-byte requests, 256-byte replies and 128 persistent connections. A 60-cell screen over work size, offered rate up to saturation and service CPU affinity (1, 2 or all logical CPUs) covered 2.52 million requests. Three points selected by a fixed rule were then confirmed with new seeds (600,000 requests). The success criterion was fixed in advance: an upper 95% bound on the CPU ratio below 0.90 with a P99 bound no worse than 1.05, or the reverse, and throughput no worse than 0.98.

CPU accounting. Process-group CPU is user plus system time charged to the service's threads, including io-wq. Softirq work charged to other contexts and whole-host totals are reported separately and never summed with it. Outputs are checked by exact replay; failures and timeouts are kept in all statistics.

4 RESULTS

4.1 LLM serving: the transport does not matter

The top half of Table 1 compares the two transports over 172,800 requests with no failures. Throughput is identical, and the differences in completion P99 (-10 ms, interval $[-64, 44]$) and process CPU cross zero. An earlier variant of the kernel path measured 18.7% lower process-group CPU than native after the polling fix in §4.3. That difference did not reproduce in the final integrated path and never appeared in whole-host CPU.

Table 2: Confirmation at 1,000 requests/s (five pairs each). Ratios are kernel-owned over native; values below 1 favor the kernel path.

Point	CPU ratio [95%]	P99 ratio [95%]
FFT, all CPUs	1.011 [0.995, 1.027]	1.254 [0.514, 3.058]
GEMM, all CPUs	1.019 [0.967, 1.074]	1.160 [0.497, 2.706]
GEMM, one CPU	1.129 [0.956, 1.334]	2.589 [1.196, 5.607]

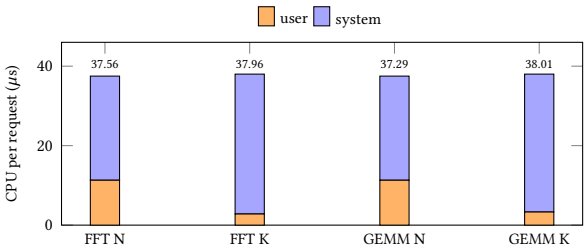


Figure 2: Process-group CPU per request at 1,000 requests/s, native (N) versus kernel-owned (K). Bars are coarse tick-based user/system splits; numbers on top are high-resolution totals. Removing the userspace loop moves work into system time.

Both resident paths had a multi-second tail while vLLM stayed near 0.6 s. The cause was capacity, not transport. With 16 slots the offered load exceeded delivery, so the queue grew for the whole run: native time to first token rose from about 235 ms in the first 10 s to 3,003 ms in the last 10 s, while generation stayed near 86 ms. Raising common capacity to 32 slots (bottom half) cut P99 from 3,337 to 217 ms, below vLLM's 406 ms at equal throughput. It also cost 0.033 ms of mean time per token, more graph and KV memory, and 0.34 J more device energy per request than vLLM.

The resident executor also used 10–16× less process CPU per request than the vLLM server. Most of that gap comes from moving step scheduling onto the GPU and from a simpler protocol that returns token IDs instead of JSON text, and both transports share it. Kernel networking does not explain it.

4.2 Small requests: no CPU or tail benefit

Short GPU work is where host overhead should matter most, yet no point met the success criterion (Table 2). The largest CPU reduction anywhere in the screen was about 5.8%, and with one logical CPU the kernel path's P99 was 2.6× worse. Whole-host CPU ratios (0.990, 0.925 and 0.988) all had intervals crossing 1. Both transports first saturated at the same offered level. Longer GPU work narrowed throughput gaps without producing a CPU advantage. All 3.12 million requests completed without an observed failure.

Figure 2 shows why. The kernel path cut user time from 11.3 to 2.8 μs per request, but system time grew from 26.2 to 35.2 μs. The totals stayed within about 1%.

4.3 Fixes that mattered more than the transport

Table 3 lists the changes with the largest measured effects. Two are general lessons for anyone writing a TCP front end for a GPU. First,

Table 3: Before/after effects of individual fixes (LLM workload). "Both" means the fix and its gain apply to both transports.

Fix	Scope	Effect
TCP_NODELAY	both	token-gap P99 40.5→2.3 ms; tokens/s 8,842→11,260
EOF reclamation	kernel	failed requests 4,157→0 of 57,600; tokens/s 7,793→11,623
Active-wait polling	kernel	process CPU 1.70→0.63 ms per request
Mixed prefill and decode [9]	both	mean time per token 1.43→1.23 ms (−14%)
16→32 GPU slots	both	completion P99 3,337→217 ms

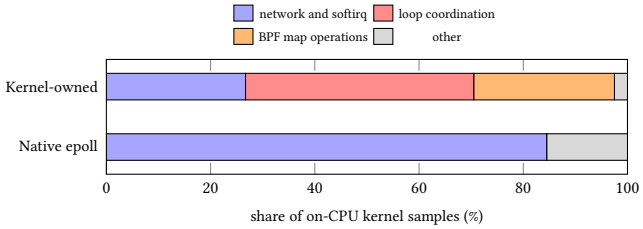


Figure 3: Where the service thread's kernel samples go (perf at 499 Hz, LLM workload at 192 requests/s, before the active-wait fix). The kernel-owned thread also had 4.1× more on-CPU samples (10,086 versus 2,455).

streamed small replies interact badly with Nagle's algorithm and delayed ACKs: a 4-byte write left 36 queued bytes waiting about 41 ms for an ACK, so every stream paid a 40 ms gap until the listener set TCP_NODELAY. Second, under sustained load the kernel path kept finished connections until an unrelated global queue drained; 2,947 sockets sat in CLOSE_WAIT, the sockhash filled, and new requests failed with E2BIG. Per-connection reclamation removed all 4,157 timeouts. A third fix removed a busy-polling branch in the BPF loop, where 71% of the loop thread's kernel samples had been coordination and BPF map operations.

5 WHY THE KERNEL PATH SAVES NO CPU

The expensive part stays. Per request, the dominant cost is TCP/IP processing: interrupts and softirqs on receive, socket buffers, the state machine, ACKs and transmission. Sockmap runs after TCP has reassembled the stream, so both transports pay for the full stack. What the kernel path removes is the system call boundary and one user copy, which is a small fraction of the roughly 38 μs each request costs on this host.

The worker moves rather than disappears. Someone must notice that the GPU has finished and submit the reply. The GPU cannot cheaply trigger kernel work, so the BPF loop runs in a service thread that waits for completion events or a 60 μs timer, reads queue state through BPF maps, and submits SQEs. Its time is charged as system time, which is exactly the shift in Figure 2. An epoll server batches many ready sockets per system call, while the BPF path pays for program invocations, map operations and SQE/CQE handling on every request.

Figure 3 makes this concrete for the LLM workload before the active-wait fix. The native thread spends 85% of its kernel samples in networking and softirq, which is the work both designs must do. The kernel-owned thread was on CPU four times as often, and 71% of its kernel samples were loop coordination and BPF map operations that the native design does not have. The fix in §4.3 removed most of this extra work for LLM serving, but in the small-request experiment the remaining coordination still cancelled the user-time savings.

Fewer cores make it worse. With one logical CPU the loop competes with network softirqs, and timer-driven wakeups add delay that readiness-driven epoll avoids, which matches the 2.6× worse P99.

When moving into the kernel does pay off. eBPF systems with large gains intercept work *before* the expensive layers: BMC [1] and Electrode [2] act at XDP [3] before the network stack, and XRP [4] resubmits storage requests from the NVMe driver. GPU networking systems that remove the CPU from the data path bypass the host stack entirely, through RDMA in GPUnet [5] or direct NIC queues in DOCA GPUNetIO [6]. Keeping ordinary Linux TCP, as BPFusion does for compatibility, means accepting the stack's cost on every request.

6 RELATED WORK

GPU-side I/O and services. GPUnet [5] gave GPU programs a socket interface backed by a CPU proxy and RDMA. The LLVM GPU libc work exposes an io_uring submission ring to device code [7], so letting a GPU program drive Linux I/O without a userspace proxy is not new. DOCA GPUNetIO [6] lets GPU kernels operate NIC queues directly. BPFusion differs in keeping the unmodified Linux TCP stack and putting connection ownership in eBPF, and our results suggest that this choice is what bounds its benefit.

In-kernel extensions. BMC [1], Electrode [2] and XRP [4] report large gains from eBPF because they act before costly stack layers. Kernel-bypass datapaths such as Demikernel [11] skip the kernel entirely. Sockmap and the io_uring BPF interfaces [12] run after TCP, which places BPFusion in a different cost regime.

LLM serving. Continuous batching [10], paged KV memory [8] and mixed prefill and decode batches [9] are the scheduling techniques that moved our tail latency. Persistent GPU kernels for inference, such as Mirage's persistent kernel [13], move step scheduling onto the GPU. BPFusion reuses existing model graphs and does not contribute model-level optimizations.

7 LIMITATIONS

All results come from one host with 24 logical CPUs and one RTX 5090, with a single remote client, small models, fixed-shape requests of at most 64 words, a binary protocol and no TLS. Process-group CPU excludes softirq work charged to other contexts, and whole-host counters include unrelated tenants and our own observer, which cost 2 to 34 μs per request. Five process pairs per point give nominal intervals rather than strong multiplicity guarantees. A

host whose cores are scarce for other reasons, a NIC with interrupt coalescing tuned for this workload, or a design that bypasses the host stack could change the conclusion; we did not test those.

8 LESSONS AND CONCLUSION

We draw four lessons. *Measure the cost floor before moving code across a boundary*: if most per-request CPU is in the TCP stack, relocating the loop cannot remove it. *Compare against a baseline that shares everything except the variable*: our same-executor epoll server separated transport effects from scheduling effects that an external server would have confounded. *Account CPU in every scope*: lower user time was offset by system time, and whole-host totals were unchanged. *In small-model serving, look at capacity and TCP small-send behavior first*: those changes moved tails by an order of magnitude, while the transport did not.

BPFusion remains useful as an open research prototype for kernel-owned TCP around reusable CUDA graphs on stock Linux, and as a starting point for designs that bypass the host stack. Source, raw data and all findings are at https://github.com/yunwei37/BPFusion (Apache-2.0).

ACKNOWLEDGMENTS

The prototype and experiments were developed with AI coding agents under the author's direction.

REFERENCES

[1] Y. Ghigoff, J. Sopena, K. Lejeune, G. Thomas. BMC: Accelerating Memcached using Safe In-kernel Caching and Pre-stack Processing. NSDI 2021.
[2] Y. Zhou et al. Electrode: Accelerating Distributed Protocols with eBPF. NSDI 2023.
[3] T. Høiland-Jørgensen et al. The eXpress Data Path: Fast Programmable Packet Processing in the Operating System Kernel. CoNEXT 2018.
[4] Y. Zhong et al. XRP: In-Kernel Storage Functions with eBPF. OSDI 2022.
[5] S. Kim et al. GPUnet: Networking Abstractions for GPU Programs. OSDI 2014.
[6] NVIDIA. DOCA GPUNetIO Architecture and Design. DOCA 3.5 documentation, 2026.
[7] LLVM Project. GPU-driven I/O with io_uring. LLVM Blog, August 2025.
[8] W. Kwon et al. Efficient Memory Management for Large Language Model Serving with PagedAttention. SOSP 2023.
[9] A. Agrawal et al. Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve. OSDI 2024.
[10] G.-I. Yu et al. Orca: A Distributed Serving System for Transformer-Based Generative Models. OSDI 2022.
[11] I. Zhang et al. The Demikernel Datapath OS Architecture for Microsecond-scale Datacenter Systems. SOSP 2021.
[12] J. Axboe. Efficient IO with io_uring. 2019.
[13] Cheng et al. Mirage Persistent Kernel. OSDI 2026.